TECHNICAL ENGLISH • MODULE 1 (EASA DOCS, SAFETY & AUDITS)

Session 1: EASA Rules & Modal Precision

Navigating Airport Regulations, Modal Authority & Safeguarding in Plain English

📌 Lesson Summary & Operational Context

In this opening session of Technical English, students learn how European aerodrome regulations are organized under **Regulation (EU) No 139/2014** and Certification Specifications (CS-ADR-DSN). We examine how modal verbs define legal obligations: **mandatory hard law (shall/must)**, **prohibitions (shall not)**, **advisory recommendations (should)**, and **permissions (may)**.

Students practice reading authentic CS-ADR-DSN Subpart H obstacle limitations, decode dense legal expressions into plain operational English, catch intentional errors in a partner listening game, and role-play a polite, professional negotiation with a construction contractor erecting a 48-metre crane near an active runway.

🎯 5 Core Spoken Learning Objectives (B2 Standards)

01	**Navigating Technical Documents:** Apply scanning techniques to locate safety-critical clauses in official manuals without getting bogged down in dense text.
02	**The Modal Authority Ladder:** Differentiate between hard law obligation (*shall/must*), prohibitions (*shall not*), advisory guidance (*should*), and permissions (*may*).
03	**Plain English Jargon Busting:** Translate dense bureaucratic boilerplate (e.g. *pursuant to, without prejudice to, derogation*) into clear, natural operational language.
04	**Active Listening & Auditory Detection:** Listen actively to partner speech and immediately interrupt with checking phrases upon hearing compliance errors.
05	**Diplomatic Negotiation Role-Play:** Balance aviation safety priorities with commercial contractor deadlines, reaching an agreed safety compromise on crane restrictions.

📄 Worksheet Index & Table of Contents

Part	Activity Title & Pedagogical Focus	Student Mode	Page
Part 1	Warm-up, Entry Diagnostic & Rules & Meaning Matrix	Pairs • Discussion & Diagnostic	2
Part 2	Reading the EASA Crane Rule (CS-ADR-DSN.H.415) & Jargon Buster	Individual / Pairs • Reading & Translation	3
Part 3	Language Clinic: Words for Rules & Direct Action Commands	Pairs • Grammar Clinic & Speaking	4
Part 4A	Spot the Safety Bug (Partner Listening Barrier Game)	Pairs (Reader & Detector) • Spoken Interaction	5
Part 4B	Speaking Role-Play: Talking to the Builder & Peer Checklist	Groups of 3 • Negotiation & Peer Evaluation	6
Part 5 & 6	Safety Check Table, Rules to Reality & Operational Notes	Class / Pairs • Decision & Written Records	7

PART 1 Activity 1: Warm-up & Entry Spot-Check

Student Instructions:

1. In pairs, discuss the 3 opening questions about reading official technical manuals in English.
2. Answer the 4 Entry Spot-Check diagnostic questions together and check your reasoning.
3. Review the Rules & Meaning Reference Table to master the difference between hard law and guidance.

💬 Pair Discussion: Opening Questions

1. Why are official aviation manuals often 300 to 500 pages long?
Talking point: Aviation safety leaves zero room for guesswork. Rules cover every weather condition, aircraft type, and accident recommendation.

2. When you need to find one specific rule in a huge PDF, what is your personal strategy?
Talking point: Using targeted search terms (Ctrl+F) for keywords like "gradient", "28 days", or scanning the Table of Contents by Subpart.

3. Have you ever had to explain a complicated rule to someone who didn't understand it?
Talking point: Explaining the safety reason behind the rule ("protects approaching planes from obstacles") rather than just saying "it's forbidden".

🔍 Airport Rules Entry Spot-Check (Diagnostic)

#	Diagnostic Question	Plain English Answer
01	What is the difference between a Law (Regulation) and a Guide (Guidance Material)?	A **Law** is 100% mandatory (legal penalties for breach). A **Guide** provides practical advice and examples to help you comply.
02	In European aviation documents, what does the word **"SHALL"** mean?	**Shall** means a compulsory legal requirement. You have no choice: it is required by law.
03	What is an **Obstacle Limitation Surface (OLS)** in simple words?	An invisible safety slope in the sky around runways where tall objects (cranes, trees, masts) are forbidden or restricted.
04	What is a **NOTAM (Notice to Airmen)**?	An urgent official alert sent to pilots informing them about temporary hazards (closed runways, crane works, unserviceable lights).

📊 Rules & Meaning: From Strict Law to Advice

Modal Verb	Legal Commitment	Regulatory Meaning	Spoken Tone
SHALL / MUST	**100% Mandatory**	Hard law. Non-compliance constitutes a safety breach or legal violation.	Direct order / firm command
SHALL NOT	**Prohibition**	Zero tolerance. Strictly forbidden under law and operational directives.	Absolute restriction
SHOULD	**Strong Advice**	Advisable best practice under AMC; alternative methods require approval.	Professional recommendation
MAY / CAN	**Permitted / Optional**	Authorized discretion or optional flexibility permitted under conditions.	Helpful guidance / option

PART 2 Activity 2: Reading the EASA Crane Rule

Student Instructions:

1. Scan the authentic EASA CS-ADR-DSN extract below to answer the 4 comprehension checks.
2. Review the Jargon Buster table and practice replacing formal legal expressions with natural plain English.

EASA CS-ADR-DSN • Subpart H – Obstacle Limitation Surfaces (Extract)

CS-ADR-DSN.H.405 Applicability
The specifications contained in this Subpart are designed to define the airspace around aerodromes to be maintained free from obstacles so as to permit the intended aeroplane operations to be conducted safely and to prevent the aerodromes from becoming unusable by the growth of obstacles.

CS-ADR-DSN.H.415 Approach Surface Specifications
(a) Applicability: An approach surface SHALL be established for each runway direction intended for landing.
(b) Characteristics: The limits shall comprise:
    (1) An inner edge located at a distance of 60 m prior to the runway threshold.
    (2) Two sides diverging uniformly at a rate of 10% (Code 3/4) from the extended runway centre line.
(d) Slope: The slope of the approach surface SHALL NOT exceed 2.0% (1:50 gradient) for precision runways.
(e) Prohibition: New objects or extensions of existing objects SHALL NOT be permitted above an approach surface except when shielded by an existing immovable object.

GM1 CS-ADR-DSN.H.415 Construction Equipment & Mobile Cranes
(b) Construction contractors intending to erect any crane exceeding 25 metres above ground level within 4 kilometres of the aerodrome reference point SHALL provide written notification to the aerodrome operator at least 28 calendar days prior to assembly.
(c) Where a temporary obstacle penetrates an established surface, the aerodrome operator SHALL request a Notice to Airmen (NOTAM), mandate low- or medium-intensity red warning lights, and establish operational restrictions.

🔍 Reading Comprehension Checks

Q1: Where does the Approach Surface start?
✓ Inner edge located 60m prior to runway threshold (H.415(b)(1)).

Q2: Maximum allowed upward slope?
✓ Shall not exceed 2.0% (1:50 gradient) for precision runways.

Q3: Are builders allowed to build above this slope?
✓ No. New objects are forbidden unless shielded by immovable structures.

Q4: How many days notice for cranes over 25m?
✓ Written notice at least 28 calendar days prior to assembly.

🗣️ Jargon Buster (Formal to Plain English)

Formal Bureaucratic English	Plain Operational English
"Pursuant to Article 4..."	**According to / Following Article 4**
"Without prejudice to Subpart C..."	**Without affecting existing rules in Subpart C**
"Shall enter into force on..."	**Becomes legally active on [date]**
"Competent authority"	**Official oversight body (DGAC/DSAC)**
"Derogation from the rule"	**An official authorized exemption**
"Repealed with effect from..."	**Officially cancelled & replaced on [date]**

PART 3 Activity 3: Language Clinic: Words for Rules

Student Instructions:

1. Complete the 5 practice sentences below by choosing the correct modal rule word.
2. In the Speaking Clinic, practice transforming heavy bureaucratic text into direct spoken commands for ground teams.

✏️ Modal Precision Drills (Choose: shall, shall not, should, may, must)

#	Rule Sentence	Correct Modal	Operational Reason
1	Contractors _____ notify the airport operator 28 days before erecting a tall crane.	**shall**	Mandatory hard law under GM1 CS-ADR-DSN.H.415(b).
2	Unshielded obstacles _____ be permitted above the approach slope.	**shall not**	Absolute safety prohibition under clause H.415(e).
3	To reduce danger, builders _____ lower the crane boom whenever winds exceed 30 knots.	**should**	Recommended precautionary best practice (AMC).
4	The airport operator _____ issue a temporary permit if all safety conditions are met.	**may**	Authorized permission and discretionary authority.
5	All airside construction workers _____ wear high-visibility vests at all times.	**must**	Mandatory basic airside safety requirement.

⚡ Speaking Clinic: Turning Heavy Words into Direct Action Commands

Written regulations use heavy abstract nouns (*implementation, notification, cessation*). On the ramp, multinational teams need clear, active verbs. Practice transforming these extracts into direct spoken instructions:

Case 1 (Heavy Written Regulation):
"Implementation of immediate containment measures shall be conducted by the operator upon the identification of a hydraulic fluid spill exceeding 5 square metres."
→ **Direct Spoken Ramp Command: "Ground staff must contain hydraulic spills over 5 m² immediately."**

Case 2 (Heavy Written Regulation):
"Notification of the competent authority regarding runway incursions should be executed by the safety manager within a timeframe not exceeding 72 hours."
→ **Direct Spoken Ramp Command: "The safety manager should notify DSAC of runway incursions within 72 hours."**

Case 3 (Heavy Written Regulation):
"Placement of rubber chocks against the main landing gear tyres shall be performed immediately upon the complete cessation of aircraft movement."
→ **Direct Spoken Ramp Command: "Handlers must chock the main wheels as soon as the plane stops."**

PART 4A Activity 4A: Spot the Safety Bug (Barrier Game)

Student Instructions:

1. Work in pairs: **Student A (Reader)** reads their official text aloud at normal speaking pace.
2. **Student B (Detector)** follows their bugged copy and must interrupt immediately using checking phrases when they hear an error.
3. Check the revealed bugs box to verify how many discrepancies were caught, then switch roles for Scenario 2.

Target Spoken Interruption Phrases (Use as soon as you hear a bug):

- "Hold on a second, did you say SHALL or SHOULD?"
- "Wait, let me double-check that number—did you say 25 or 40?"
- "Stop there—that sounds optional, but isn't it mandatory?"
- "Wait, did you say 24 hours or 72 hours?"

Scenario 1: Apron Vehicle Movement & Speed Limits (ADR.OPS.B.028)

Student A • Official Text (Read aloud):
"Drivers operating on movement areas SHALL give way to aircraft. Vehicles SHALL NOT be driven inside the aircraft safety envelope. Operators SHOULD establish maximum apron speeds not exceeding 25 km/h on roadways and 10 km/h near stands."

Student B • Bugged Copy (Listen & Catch Bugs):
"Drivers operating on movement areas MAY give way to aircraft. Vehicles SHALL be driven inside the aircraft safety envelope. Operators SHOULD establish maximum apron speeds not exceeding 40 km/h on roadways and 10 km/h near stands."

✓ Bugs to catch: 1) "may" instead of "shall"; 2) "shall be driven" instead of "shall not"; 3) "40 km/h" instead of "25 km/h".

Scenario 2: Aircraft Refuelling & Lightning (ADR.OPS.B.055)

Student A • Official Text (Read aloud):
"Refuelling with passengers on board SHALL only take place when ground staff maintain a clear exit path. All refuelling SHALL cease immediately if lightning is detected within 5 kilometres. Personnel SHOULD verify fire equipment within 30 metres."

Student B • Bugged Copy (Listen & Catch Bugs):
"Refuelling with passengers on board SHOULD only take place when ground staff maintain a clear exit path. All refuelling SHALL cease immediately if lightning is detected within 15 kilometres. Personnel MAY verify fire equipment within 30 metres."

✓ Bugs to catch: 1) "should" instead of "shall"; 2) "15 kilometres" instead of "5 km"; 3) "may verify" instead of "should verify".

Scenario 3: Occurrence Reporting Timelines (ADR.OPS.B.080)

Student A • Official Text (Read aloud):
"Aerodrome operators SHALL report any runway incursion or aircraft damage to the competent authority within 72 hours of becoming aware of the event. Reports SHALL contain factual environmental and personnel details."

Student B • Bugged Copy (Listen & Catch Bugs):
"Aerodrome operators SHALL report any runway incursion or aircraft damage to the competent authority within 24 hours of becoming aware of the event. Reports MAY contain factual environmental and personnel details."

✓ Bugs to catch: 1) "24 hours" instead of "72 hours"; 2) "may contain" instead of "shall contain".

PART 4B Activity 4B: Speaking Role-Play: Talking to the Builder

Student Instructions:

1. Form groups of 3 students: **Student A (Safety Lead)**, **Student B (Contractor)**, **Student C (Observer / Coach)**.
2. Read the scenario carefully. You have **5 minutes to prepare your arguments**.
3. Conduct the **6-minute spoken role-play** using the Useful Phrases box below.
4. Student C provides **2 minutes of feedback** using the Peer Feedback Checklist, then switch roles.

🏗️ Operational Scenario: Runway 09 Crane Safeguarding Dispute

A construction company wants to put up a **48-metre mobile crane** 1,200 metres from the Runway 09 threshold to build a new airport hotel. The crane penetrates the approach slope by **6 metres**. The contractor needs to start tomorrow because rented equipment costs €15,000/day, but they only submitted the request yesterday!

Student A: Safety Lead

- Explain 28-day notice rule was ignored.
- Explain 48m crane penetrates approach surface by 6m.
- Propose safety conditions: lower boom to 38m, night work (23h–05h), red lights, wait for NOTAM.

Student B: Contractor

- Explain delays cost €15,000 per day in rented equipment.
- Argue crane is off to the side, not on the centreline.
- Push to start immediately, but agree to safety lights & night work if forced.

Student C: Observer & Coach

- Listen to both speakers without interrupting.
- Check modal rule words: shall, must, should, may.
- Evaluate diplomatic tone and compromise.
- Give 2 positive points and 1 improvement.

💬 Useful Conversational Phrases for Diplomatic Negotiation:

- *"According to EASA safety rules, we cannot allow..."*
- *"The issue is that the crane penetrates the approach slope by 6m."*
- *"If you agree to lower the boom to 38m, we can approve a permit."*
- *"I completely understand your deadline, but flight safety is our priority."*
- *"What we can propose instead is working during night runway closure."*
- *"Let's record these agreed conditions in writing so both sides are covered."*

📋 Student C: Peer Feedback Checklist (Evaluate both partners):

- ☐ **1. Modal Accuracy:** Did Student A use mandatory vs advisory words correctly (shall vs should)?
- ☐ **2. Commercial Arguments:** Did Student B clearly present deadline and equipment costs?
- ☐ **3. Diplomatic Tone:** Was the conversation polite, professional, and non-aggressive?
- ☐ **4. Safety Compromise:** Did both sides reach a realistic compromise (night work, lower boom)?

PART 5 Activity 5: Safety Check: What Could Go Wrong?

Simple Safety Question	Agreed Class Evaluation & Operational Answer
1. What is the problem?	A contractor wants to place a 48m crane near Runway 09 with 24 hours notice, penetrating the approach surface by 6m.
2. What could happen?	An approaching aircraft in low visibility or windshear could collide with the crane, or pilots could get distracted.
3. How do we fix it?	1) Reduce crane height to 38m maximum; 2) Mandate medium-intensity flashing red lights; 3) Restrict works to night runway closure (23h00–05h00); 4) Issue official NOTAM.
4. Final Decision:	**APPROVED WITH CONDITIONS:** Contractor may proceed once the 4 protective steps are verified on site.

⏱ **Rules to Reality: 60-Second Pair Spoken Wrap-Up**

Pick one operational rule from today (crane height, apron speed, refuelling in lightning, or glycol runoff). In pairs, deliver an unscripted 60-second operational summary explaining: 1) What is the exact rule? 2) Why do ground teams sometimes ignore it? 3) What is the real-world safety consequence on the ramp?

PART 6 Activity 6: Operational Notes & Safeguarding Decision

Student Written Record: Use the lines below to record your team's role-play conclusions and safeguarding decisions.

1. Negotiation Outcome with Building Contractor:

2. Safeguarding Record & NOTAM Alert Decision: